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2026 SECONDARY 2 COMMUNICATION SCIENCE			
S/NO.: 3	TOPIC: Signals	UNIT: Wave Model - Waves Characteristics, EM Waves, Space Technology & Sound	NOTES 3

Knowledge	Cognitive Skills or Practical Skills
3.1 General wave properties	(a) Describe what is meant by wave motion as illustrated by vibrations in ropes and springs and by waves in a ripple tank. (b) Show understanding that waves transfer energy without transferring matter. (c) Define speed, frequency, wavelength, period and amplitude. (d) Recall and apply the relationship $\text{velocity} = \text{frequency} \times \text{wavelength}$ to new situations or to solve related problems. (e) Compare transverse waves and longitudinal waves and give suitable examples of each.
3.2 EM Waves	(a) Describe light and radio waves as components of the electromagnetic spectrum and recognize the other components. (b) Show an awareness that EM radiation (e.g., infrared, ultraviolet and light) has both beneficial and harmful effects. (c) State that light and radio waves are electromagnetic waves and have similar properties. (d) Recall properties of EM (in relation to Light and Radio waves) (e) Recognize range of frequencies for Light and Radio waves. (f) Recognize range of radio frequencies (e.g. Radio, Wifi, Bluetooth, mobile phones and satellite communications). (g) Understand why radios need to be tuned to the broadcast frequencies.
3.3 Space 3.3.1 Satellites & Orbits 3.3.2 Satellite Frequency 3.3.3 Types of satellites	(a) Describe how a satellite's altitude affects its orbital speed and period, and explain why Low Earth Orbit (LEO) is suitable for high-resolution imaging while Geostationary Orbit (GEO) is ideal for continuous broadcasting. (b) Compare the use of high-frequency and low-frequency electromagnetic waves in satellite communication in terms of their data transmission capacity and their vulnerability to atmospheric weather (e.g., rain fade). (c) Identify and distinguish between different types and functions of satellites based on their orbital height, data frequency and capacity (e.g., communication, weather monitoring, Global Positioning System [GPS]).
3.4 Sound 3.4.1 Characteristics of sound 3.4.2 Speed of sound and the Ear 3.4.3 Pitch and loudness 3.4.4 Microphone and Applications of sound	(a) Explain that sounds are produced due to interactions between molecules of a medium caused by a vibrating source. (b) Recognize that sound transfers energy and that it takes different lengths of time to travel from one point to another through different media. (c) Outline how the ear detects sounds in terms of the vibrations of the eardrum and ear bones, and the subsequent interpretation of sound by the brain. (d) Identify sounds of different pitch and loudness and relate the pitch to their frequencies and amplitude to loudness. (e) Describe how a basic microphone detects and converts sound into electrical signals. (f) Recognize the importance of sound in our society (e.g. for communication, music for pleasure, ultrasound in medicine) and the adverse effect of noise in our environment.
Essential Questions	1. (C) How does EM waves travel? 2. (C) What causes light to bend? F: Factual / C: Conceptual

3.1 Waves

- A wave is a phenomenon in which through or oscillations. **definition: back and forth motion**
- It is made up of **periodic motion** which is **repeated** at regular interval.

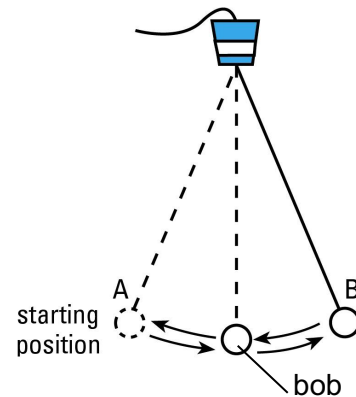
- *Example of a periodic motion:*

Swinging motion of a pendulum

The bob swings from the left (A) to the right (B) and back to its starting position (A).

One cycle (oscillation) is completed when the bob moves from A to B, and then back to A

[A → B → A].



- A wave **carries energy** **away** the **wave source**.
- The matter is **only** **displaced temporarily** and **return** to their **original equilibrium positions** (starting/ rest positions) after the wave has passed.
- A wave ... **transports energy** ... but it does ... **not transfer matter properly**
- There are **two types of waves**, transverse waves and longitudinal waves.

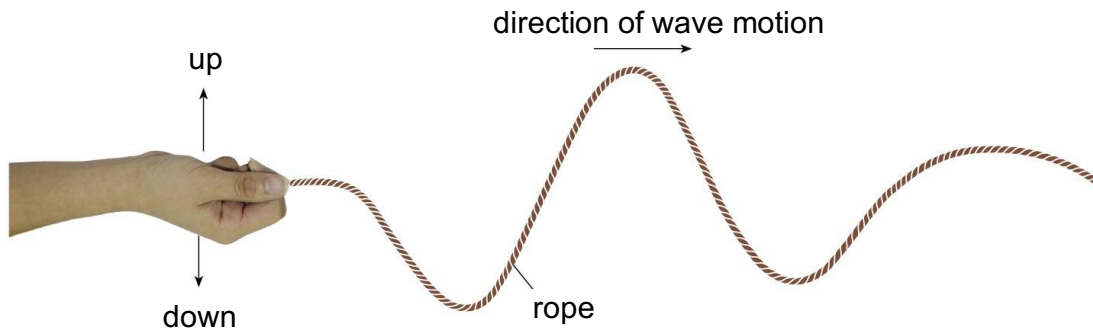
Transverse waves	Longitudinal waves
■ Transverse waves are waves which direction of vibrations are perpendicular to the direction of travel.	■ Longitudinal waves are waves which direction of vibrations are parallel to the direction of travel.
■ <i>Examples:</i> ○ Light waves ○ Radio waves	■ <i>Example:</i> ○ Sound waves

How would you classify Earthquake waves? Transverse, Longitudinal or a mix?

3.1.1 Wave Motion

Wave motion as illustrated by vibrations in a rope

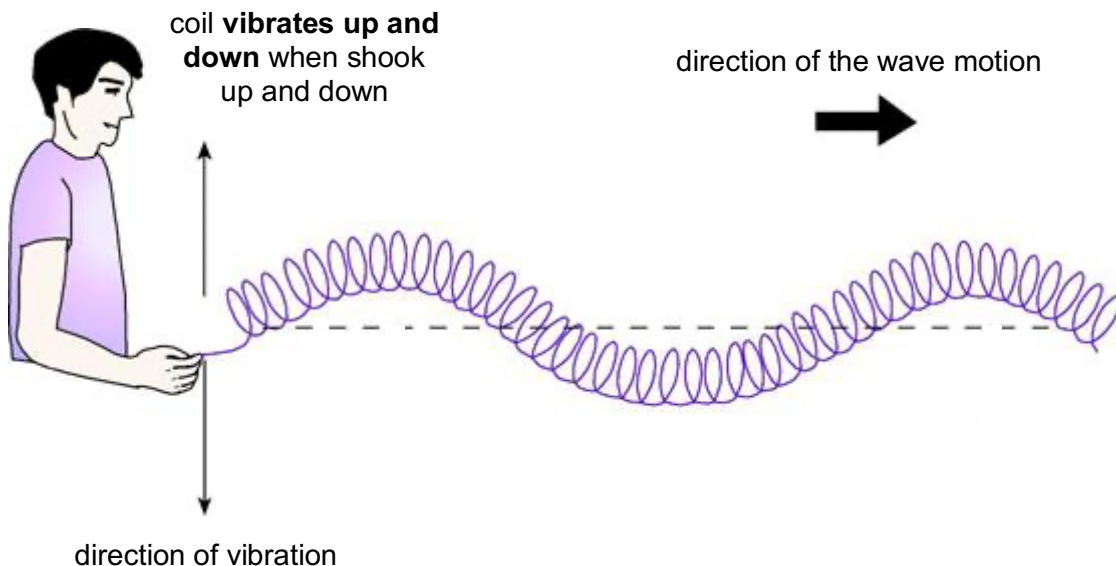
Transverse Waves



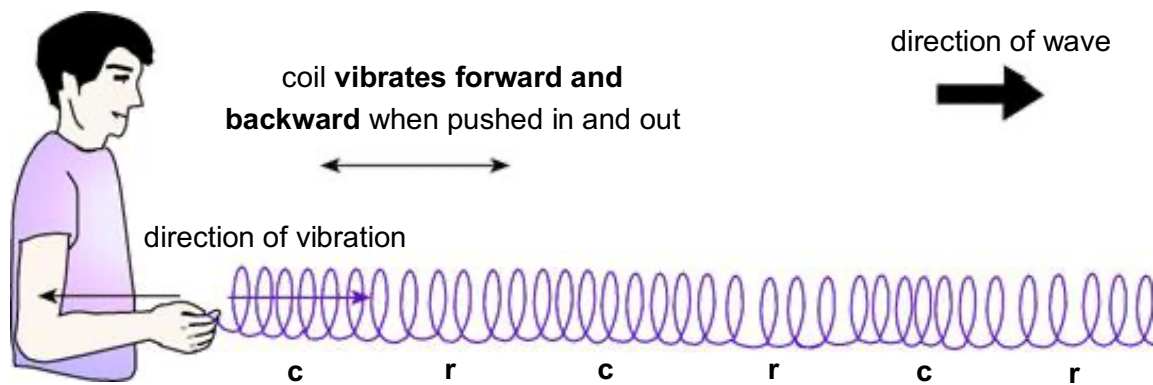
- One end of the rope is fixed and the other end is moved up and down.
- The up and down movements produce vibrations or oscillations.

Wave motion as illustrated by vibrations in a spring

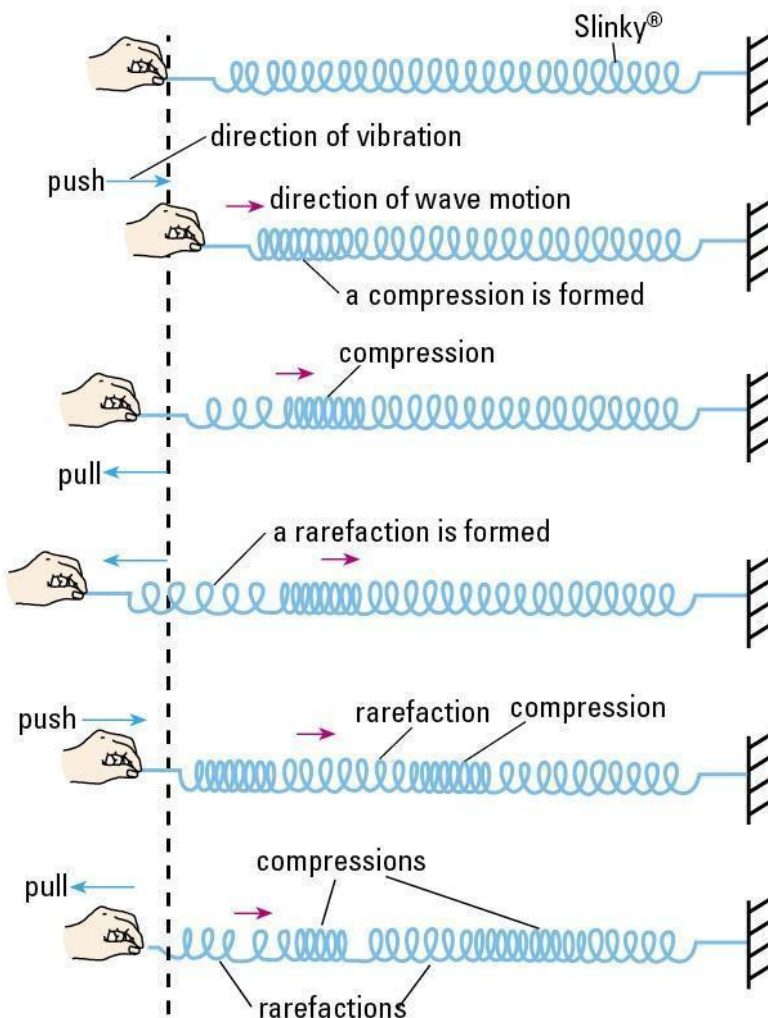
Transverse Waves



Longitudinal Waves



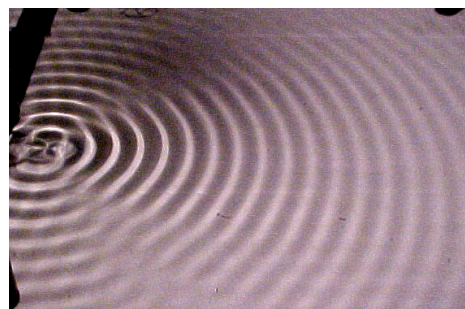
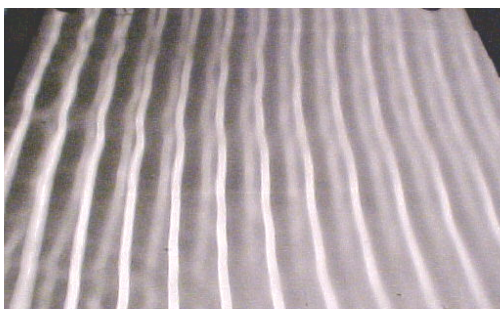
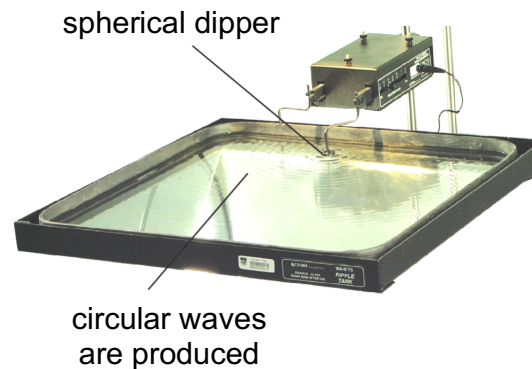
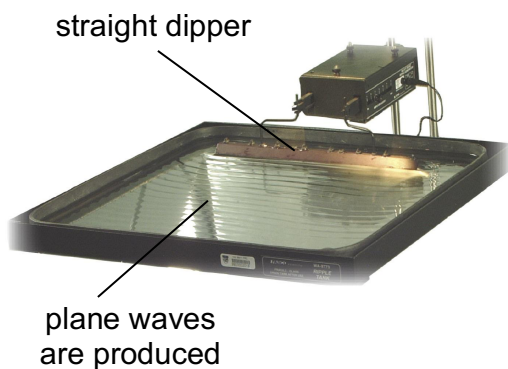
- **Compression** is part of the wave where the **particles** are *closer together*
- **Rarefaction** is part of the wave where the **particles** are *further apart*



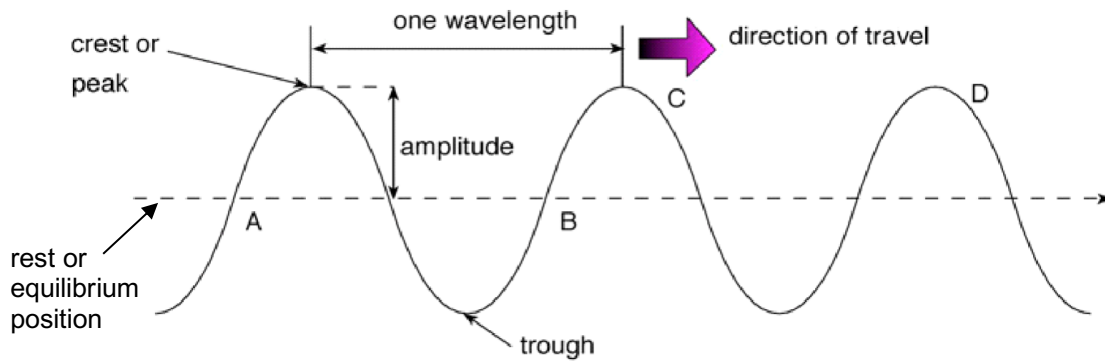
3.1.2 Wave Motion as Illustrated in a Ripple Tank

Ripple Tank

- The ripple tank consists of a shallow glass-bottom tray containing water, a dipper (wave generator), a light source situated directly above the tray and a white screen beneath the tray.
- A wave is produced as the dipper moves up and down on the water surface.
- There are two types of dipper.
 - A straight dipper is used to generate plane waves.
 - A spherical dipper is used to generate circular waves.



3.1.3 Describing Wave Motion



- **Crest** is the highest point of a transverse wave.
- **Trough** is lowest point of a transverse wave.
- Any **two points** are said to be **in phase** when they move in the **same direction**....., have the **same frequency**..... and the **same displacement**..... from the rest position.

Term	Symbol	Definition	S.I. Unit
amplitude	A	The maximum displacement of the particles from the rest position . The bigger the amplitude of the waves, the greater energy the waves carry.	metre [m]
wavelength	λ (lambda)	The shortest distance between any two points in a wave that are in phase (e.g. successive crests or troughs)	metre [m]
frequency	f	The frequency (f) of wave is the number of complete waves produced per unit time	hertz [Hz]
period	T	The period (T) of a particle in a wave is the time taken for it to complete 1 cycle .	second [s]
speed	v	The distance traveled by a wave in one second.	metre per second [m/ s]

$$\text{frequency} = 1/\text{Period}$$

Example 1

The period of a wave is 0.20 s. Calculate the frequency of the wave.

$$1/0.2 = 5.0 \text{ Hz}$$

$$\text{frequency} = \dots\dots 5.0 \text{ Hz} \dots\dots$$

Example 2

The frequency of a wave is 20 Hz. Determine the period of the wave.

$$1/20 = 0.050 \text{ (2s.f.)}$$

$$\text{period} = \dots\dots 0.050\text{s} \dots\dots$$

Speed of a wave

It shows the relationship between *speed*, *frequency* and *wavelength*.

$$\text{speed} = \text{frequency} \times \text{wavelength}$$

Example 3

A water wave has a frequency of 3.0 Hz and a wavelength of 10 cm. Calculate the speed of the wave.

$$10\text{cm} \div 100 = 0.1\text{m}$$

$$3.0\text{Hz} \times 0.1\text{m} = 0.30 \text{ m/s}$$

$$\text{speed} = \dots\dots 0.30 \text{ m/s} \dots\dots$$

Example 4

The speed of sound in water is 1500 m/s and has a frequency of 300 Hz. Calculate the wavelength of the wave.

$$f \times \lambda$$

$$1500 = 300 \times \lambda$$

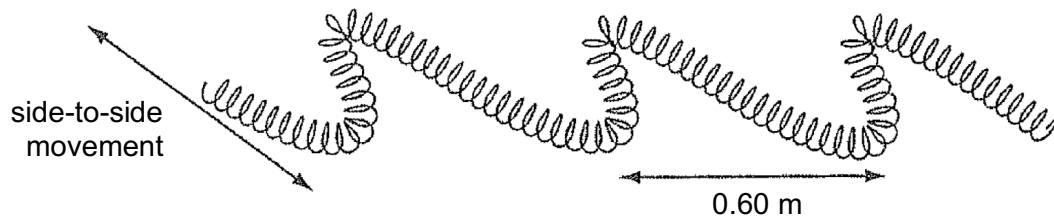
$$= 300 \times 5.0$$

$$\lambda = 5.0$$

$$\text{wavelength} = \dots\dots\dots 5.00 \lambda$$

Example 5

The diagram shows part of a spring that is shaken from side-to-side to produce a wave.



The distance between successive peaks is 0.60 m and the frequency is 2.5 Hz. Determine the time taken for the wave to travel 3.0 m along the spring.

$$2.5 \text{ Hz} \times 0.60 \text{ m} = 1.5 \text{ m/s}$$

$$3.0 \div 1.5 = 2.0 \text{ s}$$

$$\text{time} = \dots\dots\dots 2.0 \text{ s}$$

3.2.1 Properties of Electromagnetic Waves

Main Components of Electromagnetic Spectrum

V I B G Y O R

gamma rays	X-rays	ultraviolet (UV)	visible light	Infra-red (IR)	Micro-waves	radio waves
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shortest wavelength

longest wavelength

highest frequency

lowest frequency

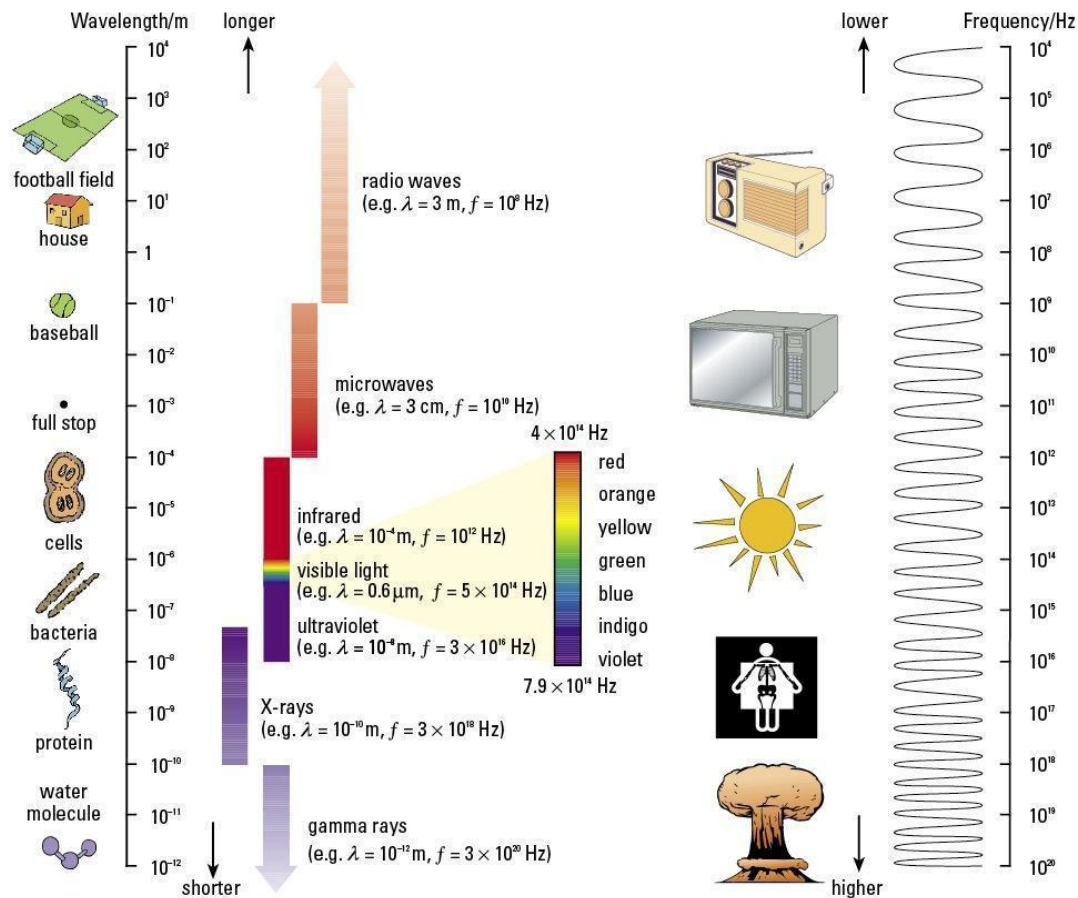
Properties of Electromagnetic Waves

1. Electromagnetic waves are **transverse waves**.
2. They transfer energy from one place to another.
3. They can travel through **vacuum**, no medium is required.
4. They travel at the **same speed** at $3.0 \times 10^8 \text{ m/s}$ in **vacuum**.
5. They exhibit wave properties such as **reflection** and **refraction**.
6. When waves travel from one medium to another, their **frequencies do not change**, only their **speed and wavelength changes**.

3.2.2 Uses of Electromagnetic Waves

Component	Uses
radio waves	radio and television communications
microwaves	microwave oven, satellite television
infra-red	infra-red remote controllers, intruder alarms, radiators (room heater),
visible light	optical fibres for medical uses and telecommunications
ultraviolet (UV) light	sunbeds, sterilisation and fluorescent tube
x-rays	x-ray radiography, security check, engineering applications (finds flaws in metal, study of crystal structures)
gamma rays	medical treatment (treatment of cancer by killing cancer cells)

- Electromagnetic spectrum is **continuous**. There are no gaps in between them.
- The **frequency** of an electromagnetic wave is related to its **energy**.
 - The **higher its frequency**, the **higher its energy**.



Example 6

The diagram shows some of the components of the electromagnetic spectrum.

γ -rays	X-rays	A	visible light	B	microwaves	radio waves
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- (a) Write down the name of component **A** and **B**.

A: ultraviolet **B:** infra-red

- (b) State the speed, in a vacuum, of visible light and radio waves.

visible light: $3.0 \times 10^8 \text{ m/s}$

radio waves: $3.0 \times 10^8 \text{ m/s}$

- (c) Red light and violet light are part of visible light.
State one difference between them.

Red light have a lower frequency than violet.

3.2.3 Radio Waves

- Radio waves have the **longest wavelength** in the electromagnetic spectrum.
- It can bend and travel around obstructions better than those of shorter wavelengths.
- Hence, Radio waves have many applications in telecommunications.

Type of radio waves	frequency	wavelength	uses
Low Frequency (LF)	30 – 300 kHz	About 10 m – 2 km	■ AM (amplitude modulated) radio. ■ submarine communication
Medium Frequency (MF)	300 kHz – 3MHz		
High Frequency (HF)	3 – 30 MHz		
Very High Frequency (VHF)	30 – 300 MHz	About 1 cm – 10 m	■ FM (frequency modulated) radio. ■ over-the-air television (terrestrial tv)
Ultra-High Frequency (UHF)	300 MHz – 3 GHz		■ terrestrial television ■ Bluetooth ■ cellular phones
Super High Frequency (SHF)	3 – 30 GHz		■ Astronomy & Satellite applications

For some useful frequencies, it is useful to know some of them.

Application	Frequency
Radio (AM)	0.5–1.7 MHz
Radio (FM)	88–108 MHz
Walkie-talkies	27 / 140 / 450 MHz
Digital TV (Ch 2–13)	54–216 MHz
Analogue TV	54–890 MHz
Digital TV (Ch 14–83)	470–890 MHz
HDTV (e.g. Channel 5 HD)	UHF (~500–700 MHz)
Mobile phones	0.7–39 GHz
Bluetooth	2.4 GHz
Wi-Fi	2.4 / 5 / 6 GHz
Satellite TV	11–13 GHz

Activity 1: Exploring the Radio Waves Around Us

Objective

In this activity, you will use a Software Defined Radio (SDR) to scan the airwaves, detect different radio signals in your surroundings, and identify what is broadcasting on them.

Equipment Checklist

Check your desk. You should have one of the following setups:

Option A (Black Device)

- [] 1 × Black FOXWEY SDR receiver
- [] 1 × Blue USB cable (USB to USB-C)
- [] 1 × Antenna

Option B (Silver Device)

- [] 1 × Silver RTL-SDR receiver
- [] 1 × USB-C adapter
- [] 1 × Antenna

 Teacher's Demo: Your teacher also has a Doorbell transmitter and a Walkie-Talkie to test later!

Setup Instructions

1. Connect the Antenna: Screw the antenna securely onto your SDR receiver.
2. Plug into your Computer: Connect the SDR receiver to your computer's USB port (use the adapter or blue cable if needed).
3. Download the software [\[here\]](#) to your desktop.
4. Launch the Software: Open the CubicSDR application on your computer.
5. Wait for devices to show Generic RTL, and then click on it and start.

Your Mission: Signal Hunting

Using the CubicSDR software, explore the frequencies around you. Your goal is to find and identify at least 10 different radio sources.

Where to look:

- Local FM Radio: Scan between 88 MHz and 108 MHz. Try to find 5 different music or news stations from Singapore, Malaysia and Indonesia!
- The Doorbell Challenge: Watch the screen while your teacher presses the doorbell transmitter. Can you spot the signal spike at 433 MHz?
- Mystery Frequencies: Look for other spikes on the screen (like the walkie-talkie). Listen in and try to figure out what they are.



Data Log

Record your findings in the table below.

SN	Frequency (MHz)	What is the source? You may google to find out what is it for that particular frequency.
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		



Discussion Question

Choose one "special" mystery frequency you found that wasn't a standard music radio station. What do you think it is used for, and why?

Activity 2: The Frequency Match Game

Objective: Students will learn how to manually tune a walkie-talkie, practice proper radio etiquette, and communicate with peers to solve a "frequency puzzle."

Duration: 20–30 minutes

Materials Needed: * One walkie-talkie per student (pre-programmed or capable of manual frequency entry).

- "Mission Envelopes" (one per student containing their assigned frequency and location).
- Copies of the activity sheet below.



Student Activity Sheet



Mission: Frequency Finder

Your Equipment: 1 Walkie-Talkie & 1 Sealed Envelope.

Instructions:

1. Open your envelope to reveal your **Assigned Frequency** and your **Starting Location**.
2. Move to your starting location and tune your walkie-talkie to your assigned frequency.
3. Broadcast your name and ask if anyone is on the line (e.g., *"This is Alex, looking for my frequency match, over?"*).
4. Once you find your match, write their name and location.
5. **The Challenge:** Now, start scanning/tuning to the *other* frequencies in the table. Try to contact as many other classmates as possible before time runs out!

Frequency Log Table

Freq (MHz)	Classmate's Name	Where are they?
430.225		
431.225		
431.525		
432.225		
432.525		
433.225		
433.525		
434.225		
434.525		
435.225		
435.525		
436.225		

Radio Etiquette Reminder

 **Remember:** Only one person can transmit on a frequency at a time. If you hear someone else talking, wait until they say **"Over"** or **"Out"** before you press your button!

3.3 Space Technology

Circular Orbits — Classroom Demonstration

Aim: Does a longer string take a longer time to complete?

What You Need

1. 1 x small pendulum bob
2. 1 x 70-cm string with a knot tied at 50 cm and 20 cm
3. 1 x Stopwatch / timer using a handphone
4. 1 x Ruler

Experiment Parameters

Parameter	String A (Short)	String B (Long)
String length (cm)	20 cm	50 cm
Number of orbits to count	10	10

Procedure

Step 1 — Short string: Tie the bob to String A (short string). Hold the free end above your head and spin the bob in a horizontal circle. Keep the radius roughly constant.

Step 2 — Measure: When the spin is steady, start the stopwatch. Count 10 complete orbits, then stop. Record the total time in the table below.

Step 3 — Long string: Repeat with String B (long string). Repeat Step 1 and 2.

Step 4 — Calculate: Divide the time taken by 10 to find the period for one orbit.

Record Your Results

String	Length	Time for 10 orbits (s)	Period per orbit (s)	Speed (Faster / Slower)
A (Short)	20 cm			
B (Long)	50 cm			

Discussion Questions

Q1: Which string had a shorter period?

.....

Q2: What does this tell us about the speed of the pendulum bob?

.....

Q3: How does length relate to Period and Speed?

Period:

Speed:

3.3.1 Satellites and Orbits

A satellite is an object in space that orbits a planet or star and remains in the gravitational field of the planet or star.

There are two types of satellites, namely natural satellites (Moon) and man-made satellites.

(a) Natural satellites

- The Moon is a natural satellite of the Earth.
- The Earth is a natural satellite of the Sun.

(b) Manmade satellites

- The International Space Station is a satellite that orbits the Earth.
- A Global positioning system is a satellite that orbits the Earth.

The orbits of satellites that move around the Earth are categorised according to orbital height (altitude).

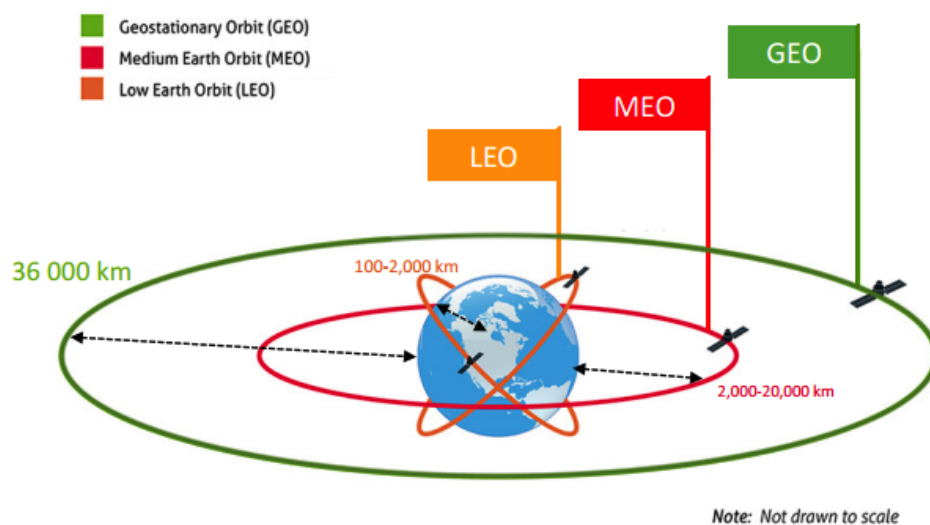


Fig 3.3.1 Three examples of satellite orbits which circle the Earth.

The following table shows the characteristics (Altitude, speed, orbital period) of different satellites, types and primary function.

Satellite / Constellation	Orbit Type	Characteristics			Primary Function
		Approx. Altitude / km	Orbital Speed / km/s	Orbital Period	
ISS (International Space Station)	LEO	420	7.66	92.8 min	Human space habitation & microgravity research
Hubble Space Telescope	LEO	535	7.60	95.2 min	Deep space optical imaging
Starlink V2	LEO	550	7.59	95.5 min	Global broadband internet transmission
Iridium NEXT	LEO	780	7.47	100.3 min	Satellite telephony & data services
GPS (NAVSTAR)	MEO	20,200	3.87	11.97 hrs	Global positioning and timing
Geostationary (GOES / Comms)	GEO	35,786	3.07	23.93 hrs	Fixed-location weather monitoring & broadcasting

Table 3.3.2: Different types of satellites at different orbits

Note:

Compare the findings from Circular Orbits — Classroom Demonstration (Page 16) and Table 3.3.2.

Period vs. Altitude Graph

Fig. 3.3.3 and Fig. 3.3.4 shows the orbital period against altitude of the satellites in Table 3.3.2, alongside a zoomed-in view of the LEO satellites.

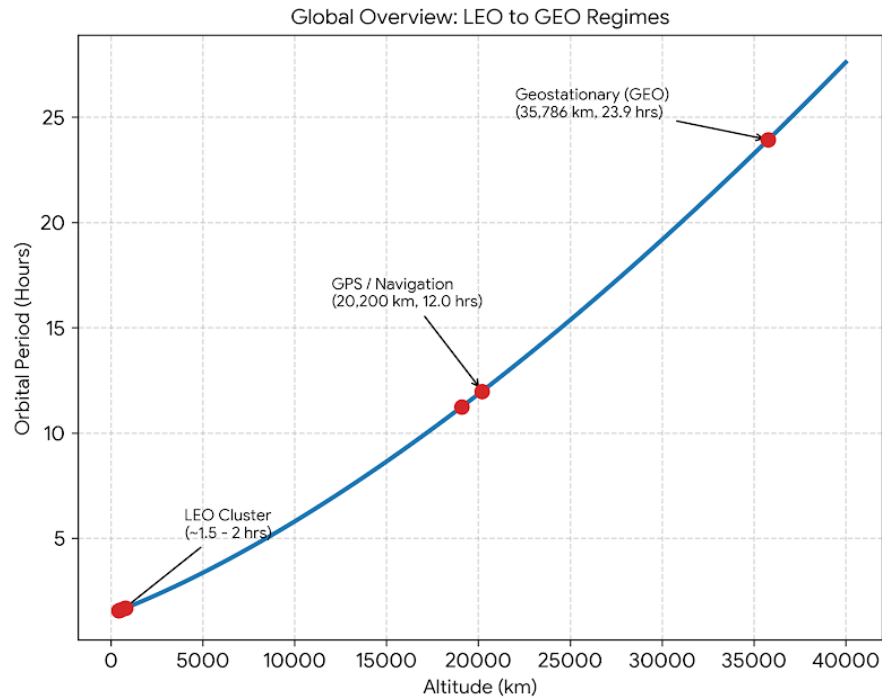


Fig. 3.3.3 Orbital period against Altitude of satellites

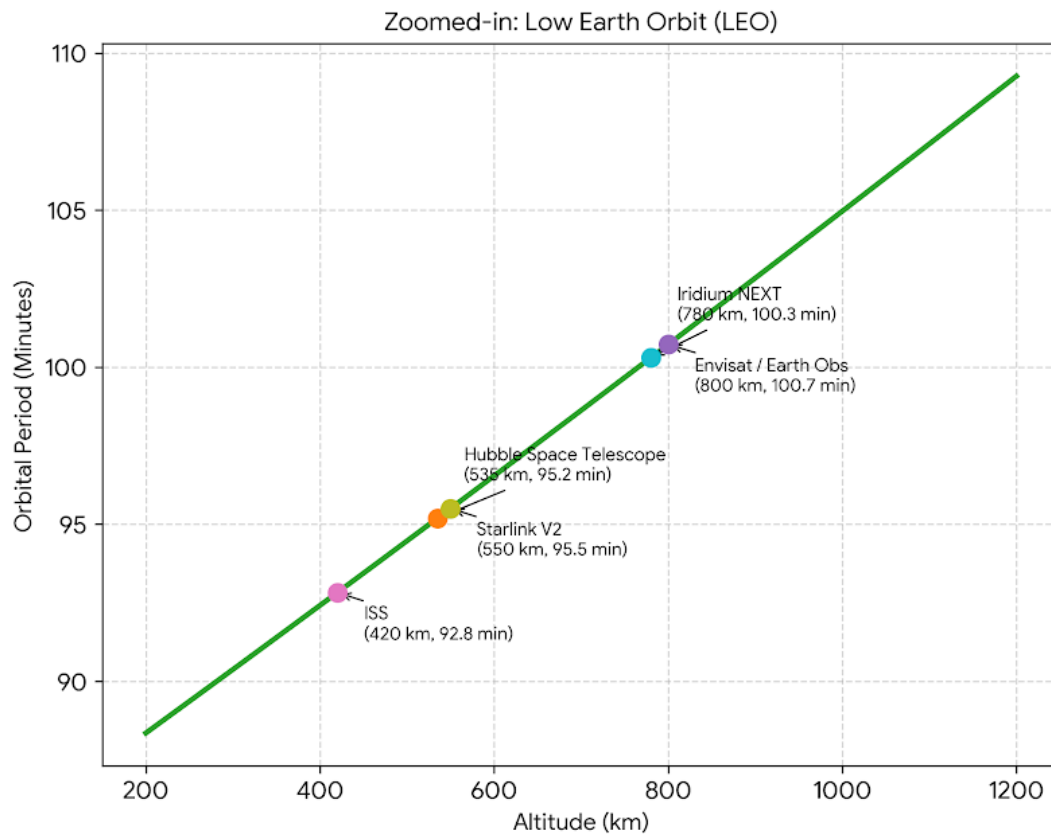


Fig. 3.3.4 Orbital period against Altitude for LEO satellites

From the both tables, what can you conclude about the relationship between the Satellite Orbital Period and Altitude?

Conclusion	The
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Speed vs. Altitude Graph

Fig. 3.3.5 shows the global curve from Low Earth Orbit (LEO) to Geostationary Orbit (GEO).

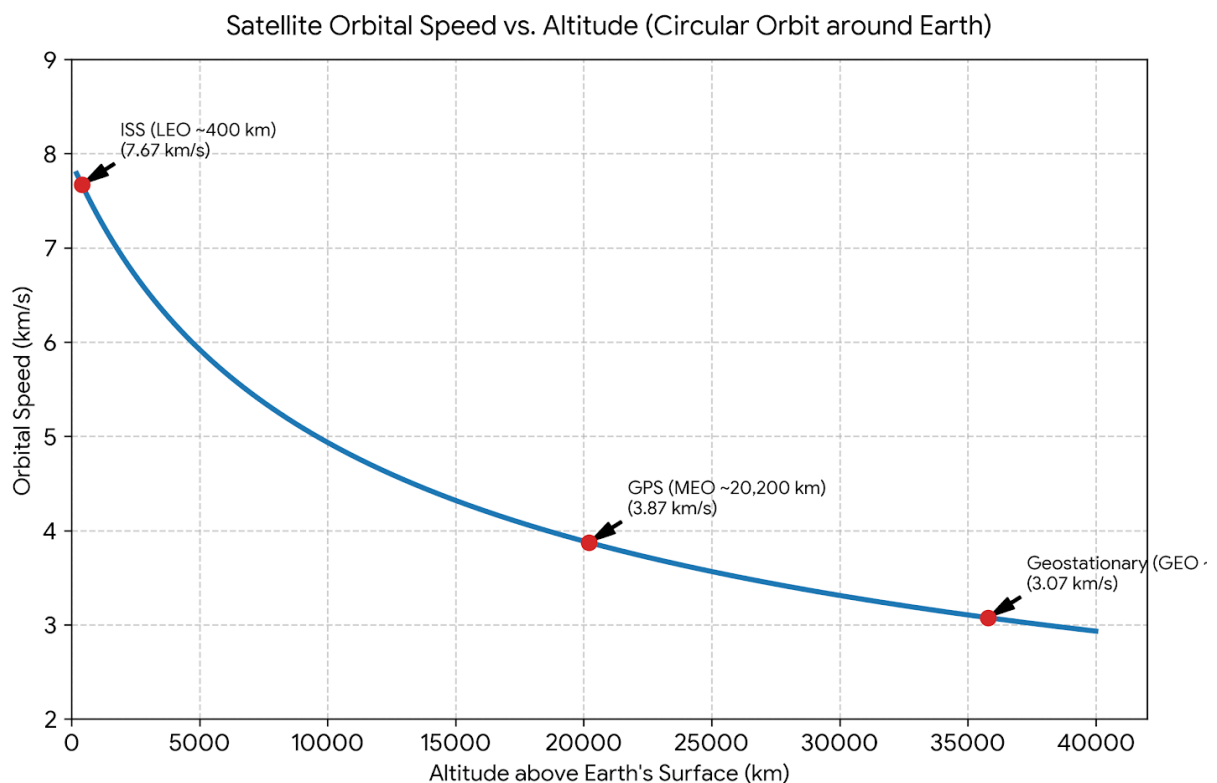


Fig. 3.3.5 Orbital speed against Altitude for satellites

What can you conclude about the relationship between the Satellite Speed and Altitude?

Conclusion	The
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Specifically, the behaviour of satellite follows the following rules:

The Altitude Rule

- **Closer to Earth:** ^{Gravity field strength} Gravity is incredibly strong. To avoid being pulled down into the atmosphere and burning up, a satellite must travel at **very high speeds**. Because its speed is fast and the path is short, its **orbital period is very short** (under 2 hours).
- **Farther from Earth:** Gravity is much weaker. The satellite travels at a **slower speed**, and because it has a massive distance to cover, its **orbital period is very long** (up to 24 hours or more).

Satellites Uses and Orbits

Figure 3.3.6 shows three satellites at different height above the earth's surface.

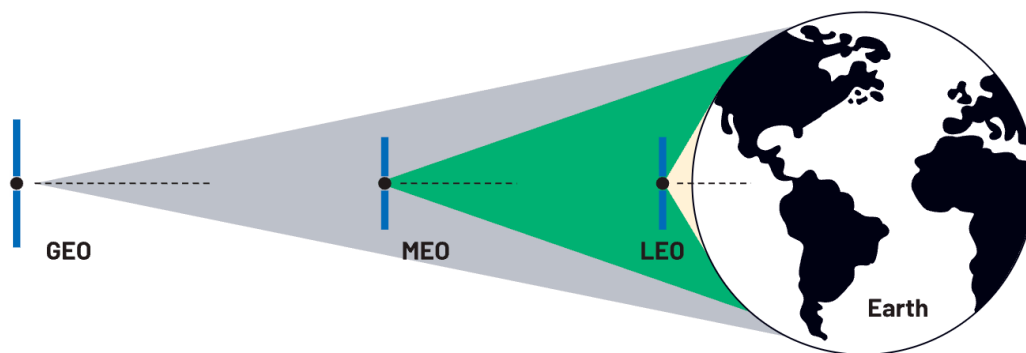


Fig. 3.3.6 Coverage of different types of satellites

What can you deduce about the relationship between the Satellite Altitude and area of coverage?

Conclusion	The Altitude low -> very high speeds -> orbital period is very short (under 2 hrs) Altitude high -> slower speed -> orbital period is very long (up to 24hours or more)
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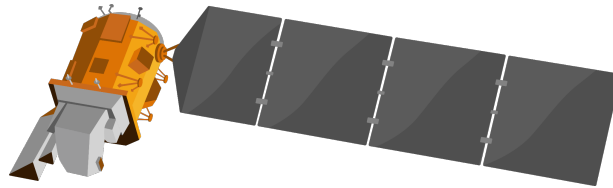
Low Earth Orbit (LEO)

- **Altitude:** 160 km to 2,000 km
- **Speed & Period:** High speed; circles Earth in about 90 minutes.
- **Why it is chosen for High-Resolution Imaging:** Being very close to the surface allows satellite cameras to capture incredibly sharp, high-resolution details of the ground (like street views or individual buildings). Because it moves around the world rapidly, it can map the entire globe over a few days.

Example: Landsat 8

Jointly operated by **NASA** and the **U.S. Geological Survey (USGS)**, Landsat 8 has been continuously mapping Earth's surface since its launch in 2013.

- **Altitude:** approx 705 km
- **Orbit Type:** Sun-synchronous, near-polar LEO
- **Speed:** Completes one full orbit roughly every 99 minutes
- **Primary Mission:** Captures high-resolution, multispectral imagery used worldwide for monitoring deforestation, urban growth, crop health, and natural disasters.



Key Onboard Imaging Sensors

- **Operational Land Imager (OLI):** Captures visible, near-infrared, and shortwave infrared light at 15-to-30-meter resolution.
- **Thermal Infrared Sensor (TIRS):** Measures surface temperatures to monitor urban heat islands and water usage.

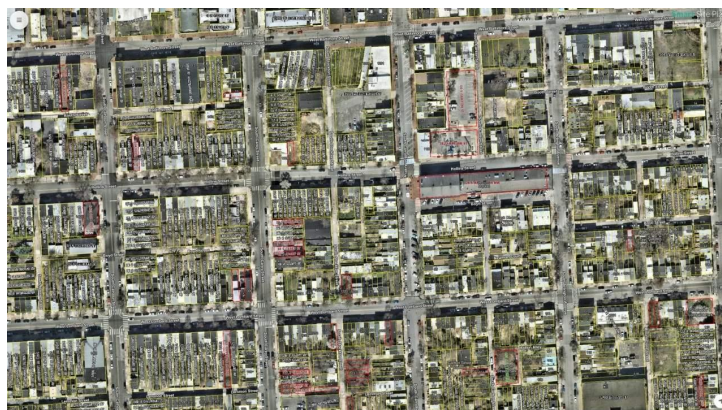


Image source: Juno Maps / NASA Worldview

Fig. 3.3.7 Satellite images

Geostationary Orbit (GEO)

- **Altitude:** 35,786 km (Much higher up!)
- **Speed & Period:** Slower speed; its orbital period is **exactly 24 hours**.
- **Why it is chosen for Continuous Broadcasting:** Because its orbital period matches Earth's rotation exactly, the satellite appears to hover over the **exact same spot** on the equator at all times. This is perfect for satellite TV and weather broadcasting because your home satellite dish can point at one fixed spot in the sky and never lose the signal.

Watch this video to visualise the Geostationary orbit:

<https://www.youtube.com/watch?v=XzhVK09YReA>

Example: ST-2

- **Operator: ST-2 Satellite Ventures** — a joint venture co-owned by **Singtel** (Singapore Telecommunications) and **Chunghwa Telecom** (Taiwan).
- **Altitude:** ~35,786 km above the Earth's equator.
- **Orbit Type: Geostationary Orbit (GEO)**, parked permanently over the **88.0°E** longitude line.
- **Speed:** 3.07 km/s. *This specific velocity allows it to keep pace with the rotation of the Earth below.*
- **Period: 23.93 hours.** Because its orbital period matches Earth's rotation, it appears stationary from the ground.
- **Primary Mission: Commercial Telecommunications & Broadcasting.** ST-2 delivers continuous Direct-to-Home (DTH) TV services, video distribution, corporate networking (VSAT), and critical maritime voice/data communications spanning **Southeast Asia, South Asia, Central Asia, the Middle East, and Mediterranean routes.**



Fig. 3.3.8 ST-2 Satellite

3.3.2 Satellites and Wave Frequency

When satellites gather information from the Earth's surface, the information needs to be transmitted to the ground. There are many different frequency bands that these information can be transmitted.

The common satellite frequency bands and its applications are as follows:

Band	Frequency Range		Typical Applications
HF	3-30 MHz		Amateur radio
VHF	30–300 MHz		Small satellites, weather satellites
UHF	300 MHz–3 GHz		Amateur radio satellites, telemetry
	L-band	1–2 GHz	GPS, satellite phones
UHF/ μ -wave	S-band	2–4 GHz	Telemetry, tracking, ISS
μ -wave	C-band	4–8 GHz	Satellite TV, communications
	X-band	8–12 GHz	Radar, military, deep space
	Ku-band	12–18 GHz	Satellite TV, broadband
	Ka-band	26–40 GHz	High-speed internet satellites

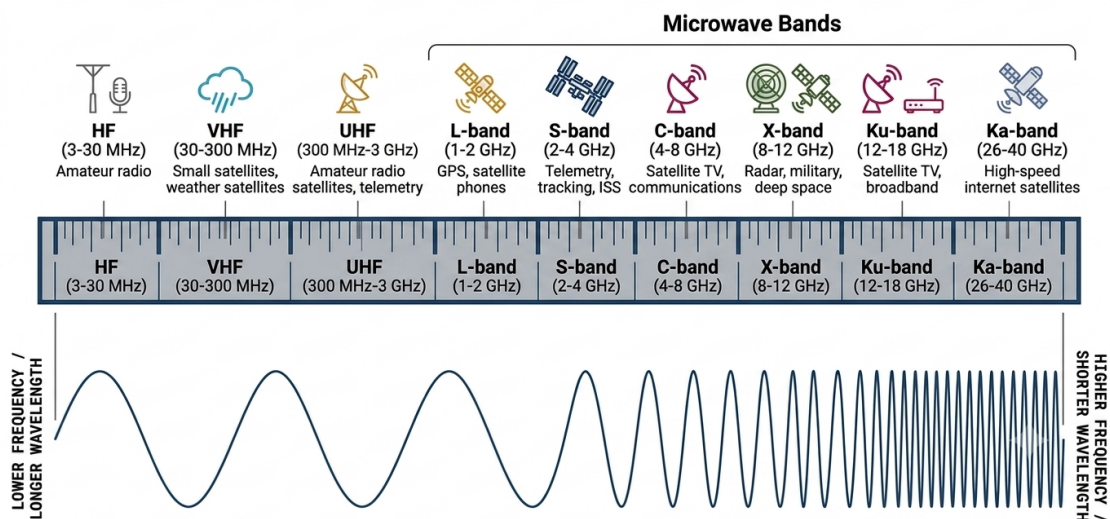


Fig. 3.3.9 Different frequency bands used by satellites

To send data through the atmosphere, satellites use different bands of the Electromagnetic (EM) Spectrum. Choosing between higher-frequency (e.g. Ka band, Ku-band) and lower-frequency EM waves (e.g., L-band, C-band) involves a trade-off between data capacity and weather resistance.

1. Data Transmission Capacity (Bandwidth)

- Higher-Frequency Waves: Have **shorter wavelengths**, meaning more wave cycles pass a point per second. This allows them to carry a much **larger volume of data (higher bandwidth)** at faster speeds.
- Lower-Frequency Waves: Have longer wavelengths and **lower data capacity**, making them **slower** at transmitting massive files but **highly reliable** for basic text or voice signals.

2. Atmospheric Penetration (Rain Fade)

- Higher-Frequency Waves: Because their wavelengths are very small, they easily collide with water droplets in the atmosphere. Heavy rain, clouds, or snow can scatter and absorb the signal, a phenomenon known as "Rain Fade."
- Lower-Frequency Waves: Their long wavelengths easily pass through or bend around atmospheric obstacles like clouds and heavy rain without significant signal loss.

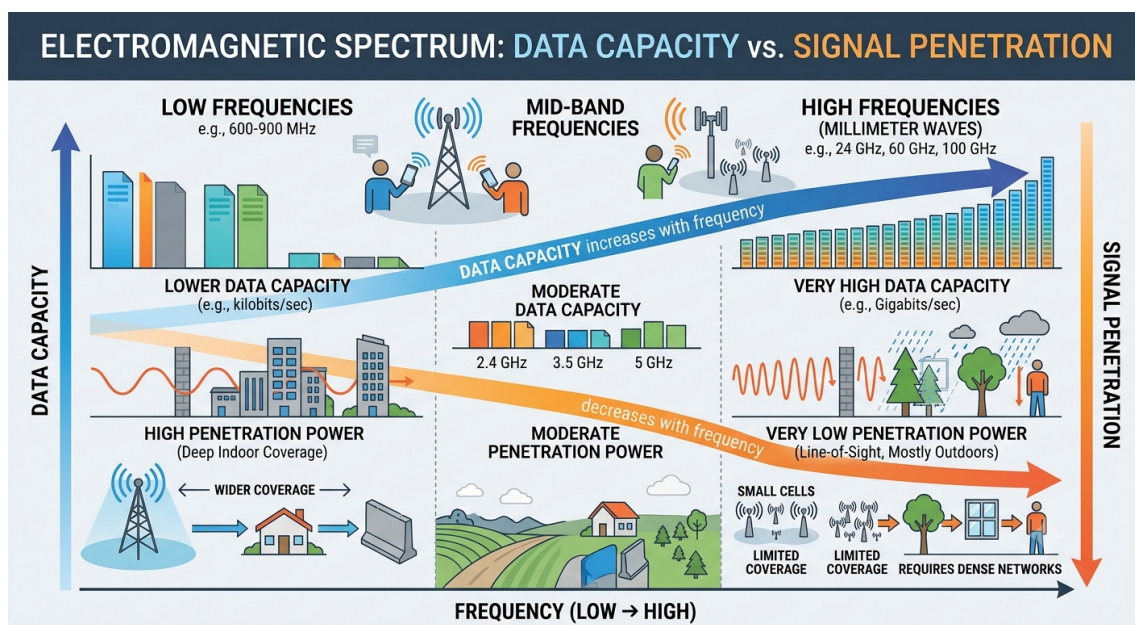


Fig. 3.3.10 The Data Capacity and Signal Penetration for different frequencies

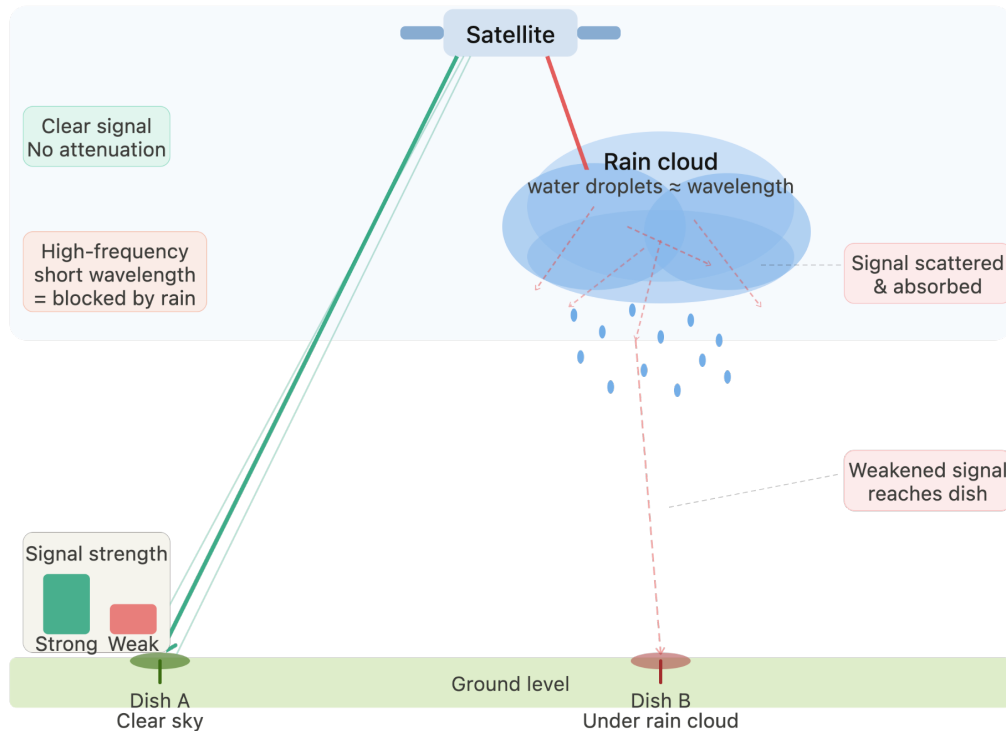


Figure 3.3.11 Rain fade on signal transmission

In summary,

Higher-frequency bands:

- **carry more data,**
- allow smaller antennas,
- but suffer more atmospheric attenuation (especially rain fade in Ku/Ka bands).

Lower-frequency bands:

- travel farther,
- penetrate atmosphere/weather better,
- but usually support lower data rates.

3.3.3 Impact of Rapid Development in Space Technology

Progress and development in space technology have increased research and development activities in various fields.

Beneficial effects	Adverse effects
Human Health Experiments can be carried out under conditions of zero gravity, for example, studies on the loss of bone tissue which causes osteoporosis.	Space Waste (Space Situation Awareness) - Space junk consists of non-functioning satellites, satellite debris, spent rockets. 95% of the thousands of man-made objects in low earth orbit (LEO) are space junk. - The increase in space junk increases the risk of collisions with satellites. This has caused satellites having to adjust their orbits several times to avoid collision with space junk.
Resonse towards climate change and disasters Provides up-to-date information about weather changes and conditions	
Innovative technology Research on growth of crystals under zero gravity	
Global education Development and progress in the field of space research	
Space Economy Economic development involving human activities in space such as space tourism	

The International Space Station

The ISS is a habitable space station that orbits the earth.

ISS is a collaboration between five space agencies, namely NASA (USA), Roscosmos (Russia), JAXA (Japan), ESA (Europe) and CSA (Canada).

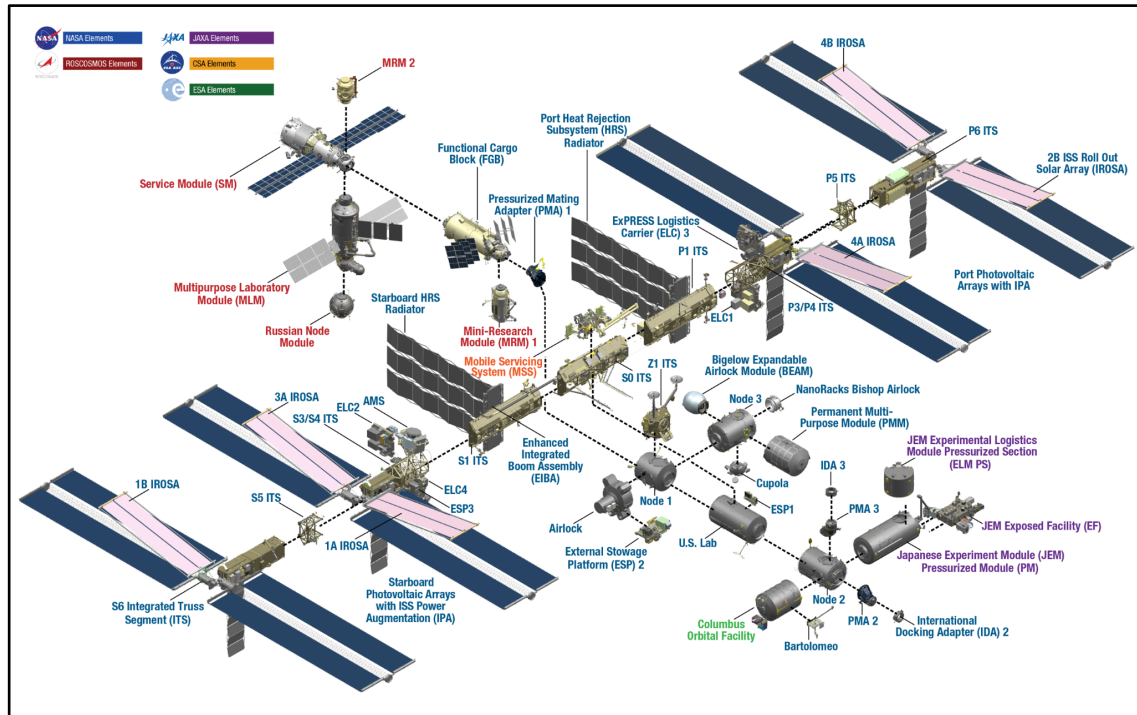


Fig. 3.3.12 International Space Station

Functions of the ISS

- Shelter for astronauts
- Place to conduct scientific research which cannot be done on Earth
- Research on the effects of microgravity on human health and the growth of microorganisms
- Centre for technological research and development of space exploration
- Monitor water resources and natural disasters on Earth

Characteristics of the ISS

- Orbits at an altitude of 408 km (Low Earth Orbit)
- Orbital speed is 7.66 km/s
- Orbital period is 92.68 minutes

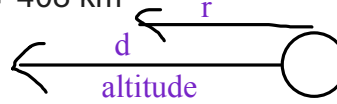
Example:

Calculate the number of orbits the ISS satellite makes around the earth a day.

Given that: **Orbit of the ISS = Radius of Earth + Altitude of satellite**

$$= 6,371 \text{ km} + 408 \text{ km}$$

$$= 6,770 \text{ km}$$



And that Period $T = 2\pi / \text{speed}$

$$2\pi r = 2\pi(6770) = 42537$$

$$42537/28000 = 1.5 \text{ hrs}$$

$$24/1.5 = 16 \text{ orbits}$$

$$7.66 \text{ km/s is}$$

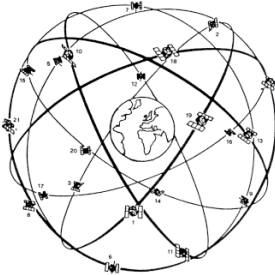
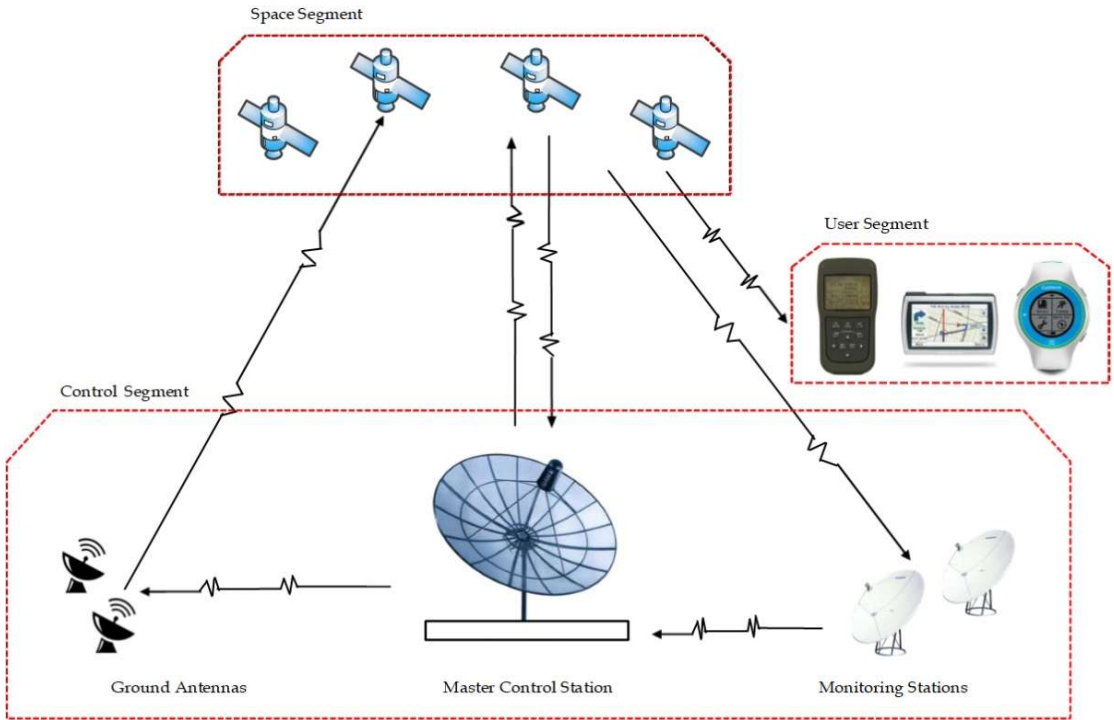
$$28000 \text{ km/hour}$$

$$7.66/1 \text{ s} = 7.66 \text{ km}/1/3600$$

$$= 28000 \text{ km/h}$$

Global Positioning System (GPS)

The GPS is a navigation system that enables users on Earth to determine the location and time with the aid of satellites.

Control segment	Space segment	User segment
Comprises a master control station, an alternate master control station, command and control antennas as well as monitor stations.	Consists of a network of 24 GPS satellites in 6 planes that orbit the Earth at an altitude of 20 000 km.	Anyone who uses a GPS receiving device or a smart phone with a GPS unit.
		
GPS control segment	Space Segment	User segment
 Fig. 3.3.13 Global Positioning System		

Watch this video about how the GPS system works:

https://www.sasbadionline.com/qrcode/wow_grafik_t5/sa_dlp/no016

The Global Positioning System, GPS is used for navigational purposes for land, sea and air transport as well as in the near space.

The location of a place on earth can be written in two formats.

- (a) Decimal degrees (DD)
- (b) Degrees, minutes (DM)
- (c) Degrees, minutes and seconds (DMS)

Activity 1

Take out your handphone and install the app phyphox. On the phyphox entry page, look out for the location (GPS) module and activate it. Take note of your current GPS location using the coordinates tab.

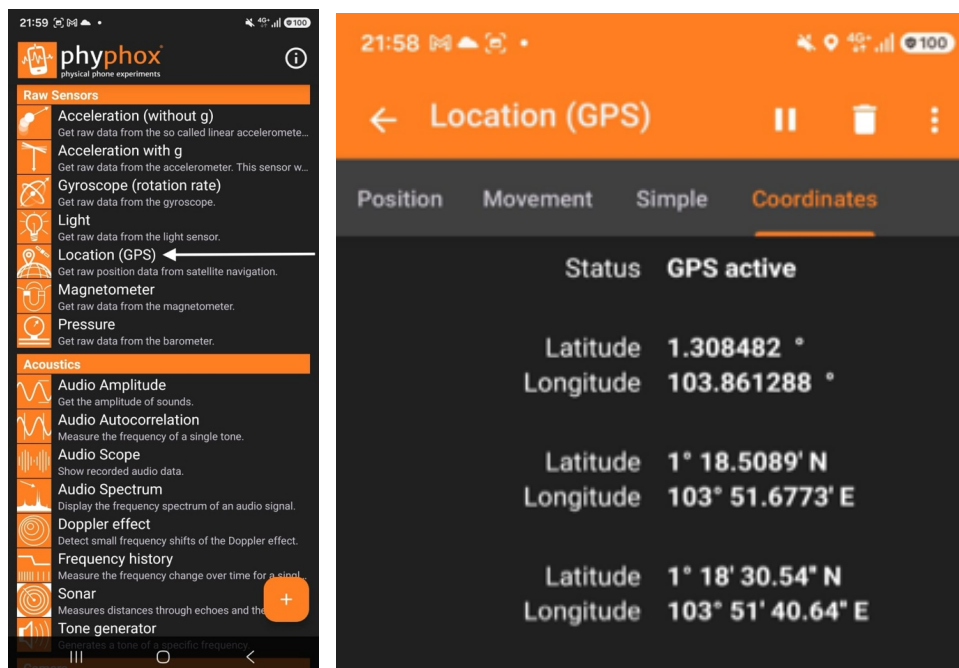


Fig. 3.3.14 phyphox app showing the location (GPS) function

Indicate the coordinates of your current location here:

	Latitude	Longitude
DD		
DM		
DMS		

Conversion of Degree to Minute to Seconds

- To convert degrees to minutes, multiply a degree by 60 to obtain minutes.
- To convert minutes to seconds, multiply a minute by 60 to obtain seconds.
- To reverse the operation, divide by 60 instead of multiplication.
- Examples 1 and 2 below shows how the different formats of coordinates are converted from one form to another.

Example 1: Convert the various coordinates from DD to DM to DMS format.

	Latitude	Longitude
DD	1.308482° (DD)	103.861288° (DD)
DM	$= 1^\circ + 0.308482^\circ$ $= 1^\circ + 0.308482 \times 60'$ $= 1^\circ 18.5089' (DM)$	$= 103^\circ + 0.861288^\circ$ $= 103^\circ + 0.861288 \times 60'$ $= 103^\circ 51.6773' (DM)$
DMS	$= 1^\circ + 18' + 0.5089'$ $= 1^\circ + 18' + 0.5089 \times 60''$ $= 1^\circ 18' 30.54'' (DMS)$	$= 103^\circ + 51' + 0.6773'$ $= 103^\circ + 51' + 0.6773 \times 60''$ $= 103^\circ 51' 40.64'' (DMS)$

Example 2: Convert the various coordinates from DMS to DM to DD format.

	Latitude	Longitude
DMS	1° 18' 30.54'' (DMS)	103° 51' 40.64'' (DMS)
DM	$= 1^\circ + 18' + 30.54/60'$ $= 1^\circ + 18' + 0.5089'$ $= 1^\circ 18.5089' (DM)$	$= 103^\circ + 51' + 40.64/60'$ $= 103^\circ + 51' + 0.6773'$ $= 103^\circ 51.6773' (DM)$
DD	$= 1^\circ + 18.5089/60^\circ$ $= 1^\circ + 0.308482^\circ$ $= 1.308482^\circ (DD)$	$= 103^\circ + 51.6773/60^\circ$ $= 103^\circ + 0.861288^\circ$ $= 103.861288^\circ (DD)$

Example 3: Carry out the conversion of your received coordinates.

	Latitude	Longitude
DD		
DM		
DMS		

Other Applications of Satellites

Applications in Environmental and Biological Monitoring

- **Tracking Wildlife:** A polar bear is fitted with a collar that records its location (**telemetry**). This data is converted onto an EM wave, sent to a satellite, and beamed to scientists to track habitat loss.
- **Climate Change:** Satellites capture radar images (**imagery**) of polar ice caps. By comparing images over time, scientists calculate the exact rate of glacial melting.

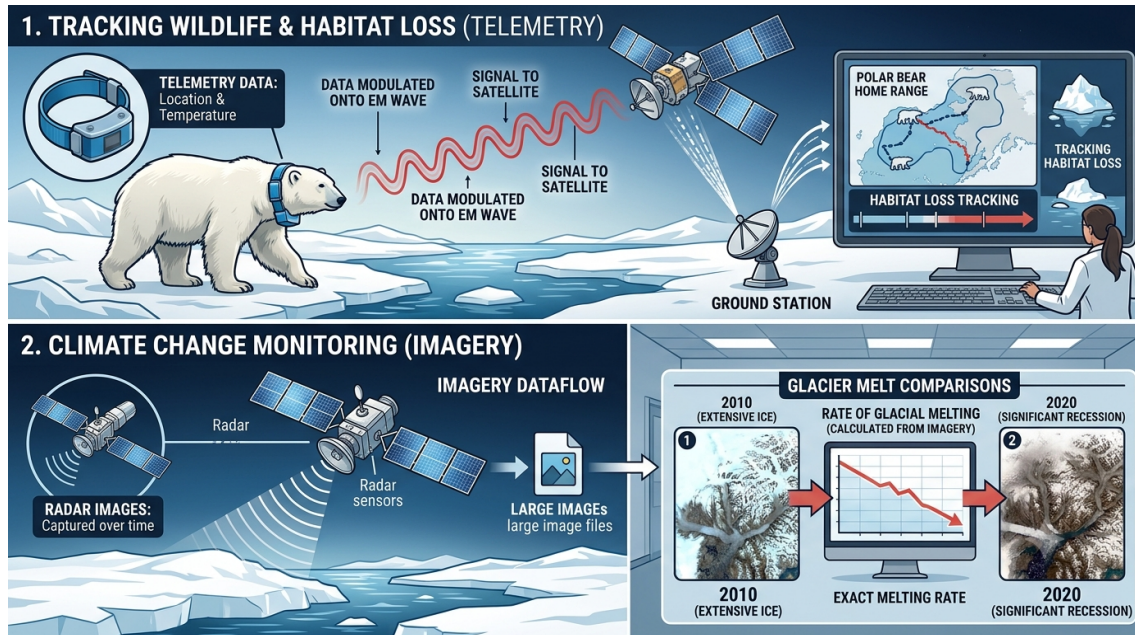


Fig. 3.3.16 Applications of satellite technology for environmental studies

THE END

3.4.1 Characteristics of Sound

- Sound is a form of energy.
- Sound waves are **produced by** a vibrating source ,
which causes a medium (material) to vibrate.
- A medium (gas, liquid or solid) is **required** to transmit sound waves.
- Sound waves are longitude waves .
 - The **direction of the vibration of the particles** is parallel to the
direction of the wave motion.

Activity 1: Experiment to prove whether sound needs a medium to travel

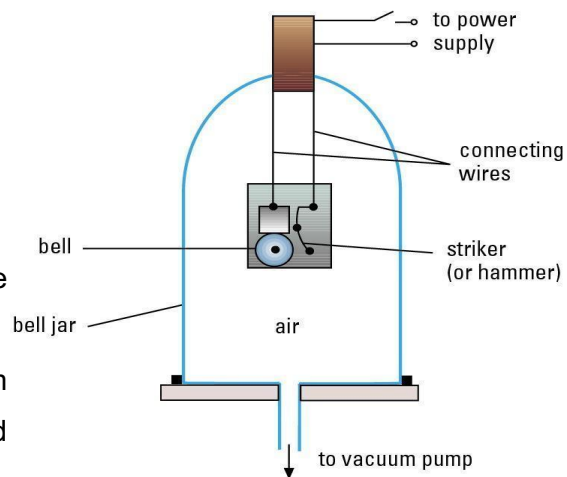
Procedure:

- A bell is suspended in a sealed jar.
- The bell is switched on.
- Air is pumped out from the jar.

Precaution:

The bell does not touch the glass and the connecting wires used are thin.

This is to prevent any sound wave from being transmitted through the glass and the wires to the outside of the jar.



Predict: What will happen if air is drawn away from the bell jar?

No sound will be produced

Observation: Watch the video <https://www.youtube.com/watch?v=hIQqX4uJtYY>

The buzzer sound gets softer. Eventually, when all the air is drawn out, there is no more sound

Explain:

The air acts as a medium for the sound to travel as sound waves. But since air is removed, there is no medium for the sound to travel and the sound gets softer.

Observations: As air is pumped out from the jar, the sound becomes fainter. Finally, the striking of the bell is seen but no sound is heard.

Explanations: Sound waves need the molecules of air (a medium) in order to travel

Watch the video on this page on how we can use a slinky coil to visualize the transmission of sound:

https://www.youtube.com/watch?v=GIkeGBXqWW0&feature=emb_logo

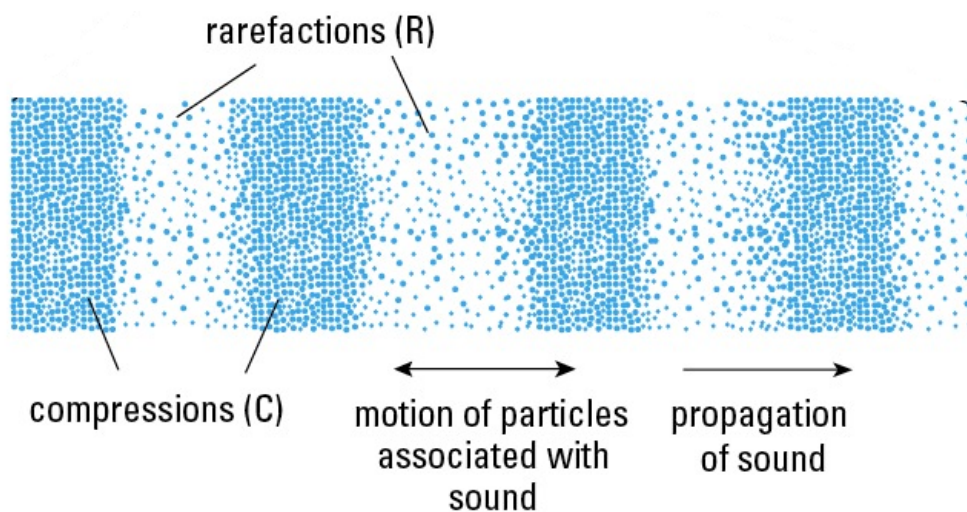
3.4.2 Transmission of Sound waves

As sound waves passes through a medium, the particles vibrate *forwards*
and *backwards* in the direction of the motion of the waves.

This creates *compressions* and *rarefactions*

Compressions are regions where the particles are *closer* *together*
and have a higher pressure than normal.

Rarefactions are regions where the particles are *further* *apart*
and have a lower pressure than normal.



3.4.3 Energy Transfers

Your teacher will now generate a sound using a musical instrument or a device such as a drum. Hold a piece of paper vertically in front of the drum and observe what happens when the drum is struck.

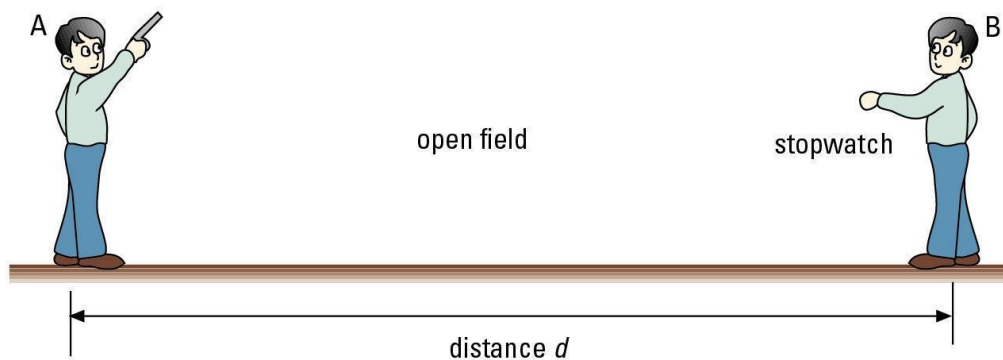
<https://www.youtube.com/watch?v=q9ezMbDpIHI> (Watch from 0:00 to 0:15 seconds)

Explain, using concepts of molecules in the air, how is the energy transferred? [3]

- When the sound is produced, the vibration of the speaker will cause the air molecules to **vibrate about their rest positions**.
- The air molecules will **vibrate and collide with their neighbouring molecules** to transfer the vibrations until the molecules at the eardrum receives it. Hence, the energy is transferred.
- The air molecules, however, do not travel from the speaker to the eardrum but just vibrates about its original position.

3.4.4 Measuring the speed of sound

Apparatus: measuring tape, stopwatch, starting pistol



Procedure:

1. Two students, A and B stands at a distance d apart.
2. Student A is holding on to a starting pistol and fires a shot into the air.
3. Student B measures the time taken from the time he sees the flash from the pistol to the moment he hears the sound.
4. He then calculates the speed by using the formula:
Speed of sound = distance / time taken.
5. The experiment is repeated, and the speed of sound determined again.
6. The average speed is obtained by taking the average in steps 4 and 5.

State a source of error in this experimental setup and suggest a method of improvement. [2]

.....

.....

3.4.5 Speed of sound in different medium

There are more accurate methods of to determine the speed of sound, and Scientists have measured the speed of sound in gas, liquid and solid.

The speed of sound generally differs in gases (air), liquids and solids as shown below:

	Gas (air)	Liquid (water)	Solid (concrete)
Speed of sound	330 m/s	1400 m/s	5000 m/s

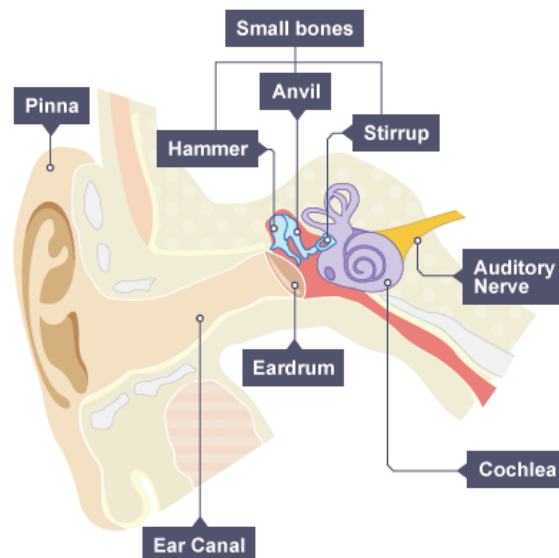
Explain why the speed of sound differs in the 3 states of matter.

[2]

The molecules in the solid are closer together. Hence, the vibrations and collisions of the molecules in a solid can occur more frequently. Sound energy can be transferred faster in solids than liquids and gases.

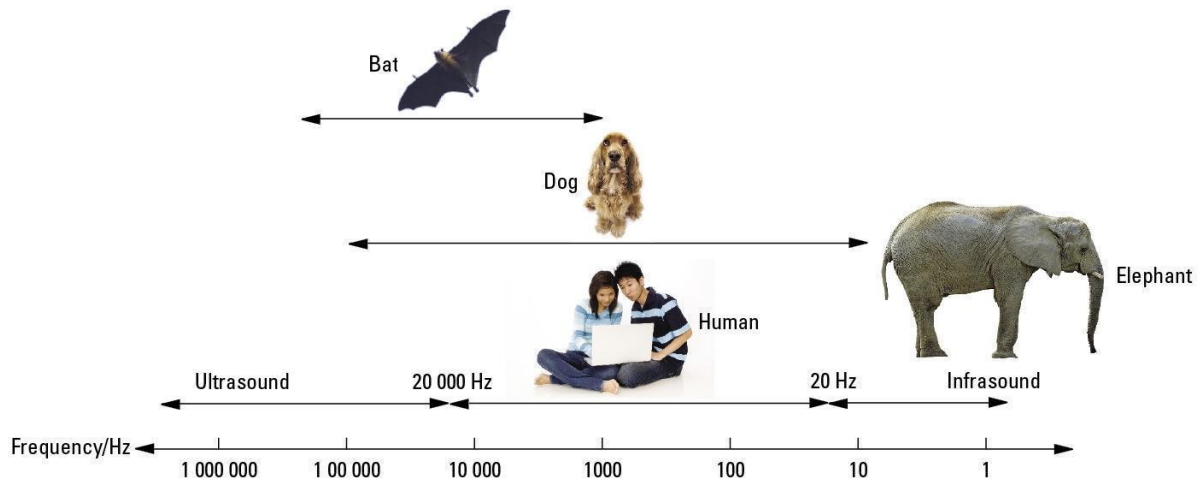
3.4.6 The Ear

We can detect sound using our ears. An ear has an eardrum inside, connected to three small bones. The vibrations in the air make the **eardrum vibrate**, and these vibrations are passed through the **three small bones (called ossicles)** to a spiral structure called the **cochlea**. Signals are passed from the cochlea to the brain through the **auditory nerve**, and our **brain interprets these signals as sound**.



A diagram to show the inner structure of the ear

- Human ear **can** detect sound with frequency from **20 Hz to 20 000 Hz**.
- This range of frequency is known as the **range of audibility**.
- Ultrasound** is sound with frequency **above** the human range of audibility.
- Ultrasound has a **frequency** **>20kHz**



Activity 3: Finding your audible frequency limits

This activity is interesting for the student as they realise the limits of their hearing. Watch the Youtube video from 0:00 to 3:30 and find out what is the upper limit of your hearing. https://www.youtube.com/watch?v=_SHFwmPQ_rQ

My upper limit is Hz and my lower limit is Hz.

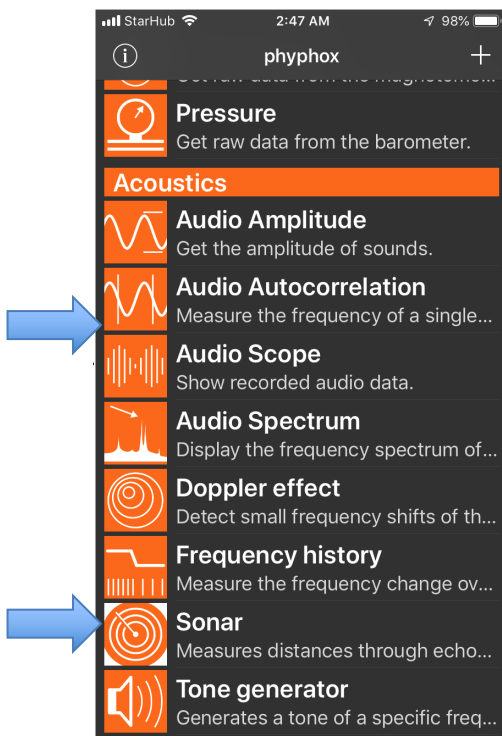
3.4.7 Pitch and Loudness

Based on your experience, you should be able to differentiate between LOUD and SOFT.

You should also have experienced hearing a HIGH PITCH and LOW PITCH.

For Activity 4A below, you can use any of the following software:

1. Audio Explorer:
2. Audacity:
3. Tuner T1: <http://bit.ly/tun-t1>

Activity 1: Investigating the relationship between Loudness and Amplitude

Phyphox is an app that can be downloaded and installed on the phone. It is an open-source software created by the RWTHAACHEN university in Germany.

Get into a group of 2 students in an isolated room. Student A will activate the **Tone Generator** and sets a frequency (e.g. 800 Hz). Student B will switch on the **Audio Scope** and place the hand phone at 1 cm away.

When student A activates the PLAY button, students B will observe the waveform of the tone generated at 1 cm away. Take a screen shot of the wave form.

Student B will observe the waveform when the hand phone is moved further from the signal source to about 5 meters away. Take a screen shot of the wave form.

Record the waveforms in the space provided below:

Distance	1 cm away	5 meters away
Wave forms		

Conclusions:

- The **loudness** of a sound depends on its
 - The the sound, the **greater the amplitude**.
 - The **softer** the sound, the **smaller the amplitude**.

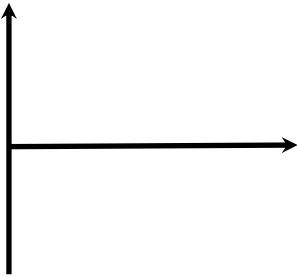
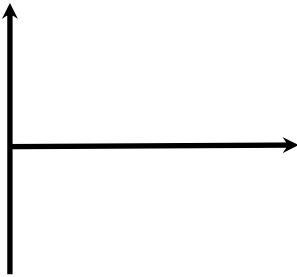
Activity 2: Investigating the relationship between Pitch and Frequency

Student A will activate the **Tone Generator** and sets a frequency of **high pitch** (e.g. 800 Hz). Student B will switch on the **Audio Scope** and place the hand phone at 1 cm away.

When student A activates the PLAY button, students B will observe the waveform of the tone generated at 1 cm away. Take a screen shot of the wave form.

Student A will now change the frequency of a **lower pitch** (e.g. 400 Hz). Student B will take a screen shot of the wave form.

Record the waveforms in the space provided below:

Frequency	800 Hz	400 Hz
Wave forms		

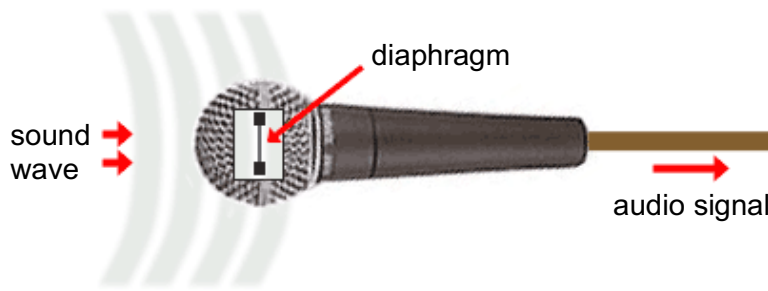
Conclusions:

- The **pitch** of a sound depends on its
 - The the pitch, the **higher the frequency, f** .
 - The **lower** the pitch, the **lower the frequency**.
 - Since frequency, **f** is inversely proportional to period, **T** , pitch is inversely proportional to period, **T ($f = 1 / T$)**.

3.4.8 Microphones

Microphones are a type of **transducer**. A transducer is a device which **transforms a signal from one energy** to another energy. Microphones transform **sound** (sound waves) into **electrical signals** (the audio signal).

A **diaphragm** is found in a microphone. It is a thin piece of material (such as paper, plastic or aluminum) which **vibrates** when it is struck by sound waves. In a typical hand-held microphone like, the diaphragm is located in the head of the microphone.



When the diaphragm vibrates, it causes other components in the microphone to vibrate. These **vibrations** are transformed into an **electrical current** which becomes the audio signal. At the other end of the audio chain, the **loudspeaker** is also a transducer which it transform the **electrical signal** back into **sound energy**.

3.4.9 Applications of Sound

How to see with sound – Jacques S. Abramowicz
<https://www.youtube.com/watch?v=4JLNb8-LOB0>

- **Communication**

Without sound waves, human beings could not communicate verbally. Your vocal cords generate sound waves that are then transmitted through the air to the ears of listeners. Modern communication technologies, such as radios and televisions, use this same basic concept to transmit sound to your ears.

- **Ocean Exploration**

Scientists use sound waves in sonar devices when they explore the oceans. Sonar sends out sound waves, which then bounce back to the source when they hit an object. Scientists can use this echo to determine the size and distance of the object that bounced the sound waves back. Navy vessels also use sonar technology to search for enemy submarines.

- **Underground Resources**

Geologists use sound waves to search for resources such as oil under the earth. They bounce sound waves into the ground and measure the way in which they travel through the earth. By measuring the ways in which the sound waves travel through the earth, geologists can make inferences about the density and makeup of the ground. Geologists can also use the waves that earthquakes produce to study the ground in a similar manner, as well as to study the effects and intensity of the earthquakes themselves.

- **Hunting in bats**

Many creatures use sound waves to hunt for food. Bats especially use a form of sonar to search for prey. Bats project sound waves that bounce off prey. When the sound waves return back to the bats, they can determine their distance from their prey. In this way, bats can effectively hunt at night despite having relatively poor eyesight. Some sea creatures, such as dolphins, use similar forms of echolocation to both hunt for prey and communicate with one another.

(Source: <https://sciencing.com/importance-sound-waves-8694751.html>)

- **Music for pleasure**

In theatres and home theatre systems, high-quality sound contributes more to overall enjoyment of the movie or event. Having surround sound helps you hear the dialog perfectly while still enjoying music and other sounds at a normal level.

(Source: <http://www.flanners.com/blog/the-importance-of-surround-sound>)

- **Ultrasound in medicine**

Ultrasound is used to examine the development of a foetus / liver.



A transmitter sends pulses of ultrasound into the mother's body. The transmitter also acts as a detector and picks up pulses reflected from the foetus and different layers of tissues in the body. The signals are processed by the computer, which puts an image on the screen. Using ultrasound is safer than using X-rays because X-rays can cause damage living cells and tissues.

Adverse effect of sound on our environment

- Noise pollution is affecting a range of animals across multiple habitats.
- Animals are altering their natural behaviours or relocating to avoid noisy areas.
- Changes in animal behaviour can have flow-on effects for whole ecosystems.
- Marine animals are also affected by noise from a range of human activities including commercial vessel traffic, oil and gas exploration, seismic surveys and military sonar.

(Source: <https://www.science.org.au/curious/earth-environment/noise-pollution-and-environment>)

Adverse effect of sound on human beings

- Hearing Problems
- Poor Cognitive Function
- Cardiovascular Issues
- Sleep Disturbances
- Trouble Communicating
- Mental Health Problems

(Source: <https://helpsavenature.com/noise-pollution-effects>)

THE END